الطالب الأكاديمي
THE ACADEMIC STUDENT

Faculty of Business Studies

BUS102

Introduction to Statistics

MTA Examination / Version C

2025-2026 / Fall

6-December-2025

Number of Exam Pages: 6 (including this cover sheet)	**Time Allowed: 1.5 Hours**

Instructions:

- Take **FIVE (5)** minutes to read all instructions carefully before attempting to write.
- Mobile phones must be in "SWITCHED **OFF**" mode throughout the examination.
- This is a **CLOSED** book examination. NO documents of any kind are allowed.
- You **MUST** answer the questions using only the answer booklets provided
- You **MUST** return the examination paper together with your answer booklets.
- Do **NOT** write your answers in pencil except graphs/sketches
- Students are **NOT** allowed to use any type of dictionary or mobile phones during the examinatic
- Students are allowed to use ***non-programmable*** calculators only (if needed).
- Any form of cheating will **NOT** be tolerated and will subject you to AOU Cheating Policy.

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Question 1: (10 marks)

1. Which scale type allows for true zero and ratio comparison?

A) Nominal
B) Ordinal
C) Interval
D) Ratio

2. Which error is due to the respondent misunderstanding the question?

A) Sampling error
B) Nonresponse error
C) Coverage error
D) Measurement error

3. What type of variable is 'eye color"?

A) Categorical
B) Ordinal
C) Discrete
D) Ratio

4. Which type of chart is best used to visualize the frequency of categories in a single categorical variable?

A) Histogram
B) Scatter Plot
C) Bar Chart
D) Polygon

5. Which of these best visualizes the five-number summary?

A) Histogram
B) Boxplot
C) Pie chart
D) Scatter plot

6. Which measure is the square root of variance?

A) Median
B) Mean
C) Standard deviation
D) Mode

7. When two dice are rolled, what is the probability that the sum is 7?

A) $\frac{1}{6}$
B) $\frac{1}{12}$
C) $\frac{1}{36}$
D) $\frac{1}{3}$

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8. What is the complement of the event 'rolling a 6 on a die'?

A) Rolling a 1

B) Rolling a number less than or equal 6

C) Rolling a number other than 6

D) Rolling a 5

9. An event with a 100% probability is called:

A) Impossible event

B) Certain event

C) Likely event

D) Rare event

10. Which of the following cannot be a probability?

A) 0.75

B) 1

C) -0.2

D) 0

Question 2: (10 marks)

A. Based in the following frequency distribution for a student exam scores: (5 marks)

Score Range	Frequency	Relative frequency	Cumulative frequency	Cumulative frequency percentage
0 to 20	2			
21 to 40	6			
41 to 60	10			
61 to 80	8			
81 to 100	4			

Calculate:

a. Relative frequency (1 marks)
b. Cumulative frequency (2 marks)
c. Cumulative frequency percentage (2 marks)

B. For the following data: 40, 20, 50, 35, 25

Find Interquartile Range (IQR), and **draw the box plot for the given data** (5 marks)

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Question 3: (10 marks)

A. In a survey, 8 people reported how many books they read last month: 2, 4, 4, 4, 5, 5, 7, and 9. What is the sample variance, sample standard deviation, and coefficient of variation for books read? (5 marks)

B. The following table is a result of a survey about the preferred mode of transportation by City

City	Car	Bicycle	Bus	Total
City A	60	25	15	100
City B	30	40	30	100
City C	20	15	65	100
Total	110	80	110	300

Find:

(5 marks)

a) What is the probability that a randomly selected person uses a bus?
b) What is the probability that a randomly selected person is from City A?
c) What is the probability that a person is from City C and uses the bus?
d) If a person is from City B, what is the probability they use a bicycle?
e) If a person uses a car, what is the probability they are from City A?

End of questions

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FORMULAS

Sample Mean

$$\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n} \quad (3.1)$$

Median

$$\text{Median} = \frac{n+1}{2} \text{ ranked value} \quad (3.2)$$

Geometric Mean

$$\bar{X}_G = (X_1 \times X_2 \times \cdots \times X_n)^{1/n} \quad (3.3)$$

Geometric Mean Rate of Return

$$\bar{R}_G = [(1 + R_1) \times (1 + R_2) \times \cdots \times (1 + R_n)]^{1/n} - 1 \quad (3.4)$$

Range

$$\text{Range} = X_{\text{largest}} - X_{\text{smallest}} \quad (3.5)$$

Sample Variance

$$S^2 = \frac{\sum_{i=1}^{n} (X_i - \bar{X})^2}{n-1} \quad (3.6)$$

Sample Standard Deviation

$$S = \sqrt{S^2} = \sqrt{\frac{\sum_{i=1}^{n} (X_i - \bar{X})^2}{n-1}} \quad (3.7)$$

Coefficient of Variation

$$CV = \left(\frac{S}{\bar{X}}\right)100\% \quad (3.8)$$

Z Score

$$Z = \frac{X - \bar{X}}{S} \quad (3.9)$$

First Quartile, Q_1

$$Q_1 = \frac{n+1}{4} \text{ ranked value}$$

Third Quartile, Q_3

$$Q_3 = \frac{3(n+1)}{4} \text{ ranked value}$$

Interquartile Range

$$\text{Interquartile range} = Q_3 - Q_1$$

Population Mean

$$\mu = \frac{\sum_{i=1}^{N} X_i}{N}$$

Population Variance

$$\sigma^2 = \frac{\sum_{i=1}^{N} (X_i - \mu)^2}{N}$$

Population Standard Deviation

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N} (X_i - \mu)^2}{N}}$$

Sample Covariance

$$\text{cov}(X, Y) = \frac{\sum_{i=1}^{n} (X_i - \bar{X})(Y_i - \bar{Y})}{n-1}$$

Sample Coefficient of Correlation

$$r = \frac{\text{cov}(X, Y)}{S_X S_Y}$$

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Probability of Occurrence

$$\text{Probability of occurrence} = \frac{X}{T} \quad (4.1)$$

Marginal Probability

$$P(A) = P(A \text{ and } B_1) + P(A \text{ and } B_2) + \cdots + P(A \text{ and } B_k) \quad (4.2)$$

General Addition Rule

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B) \quad (4.3)$$

Conditional Probability

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)} \quad (4.4a)$$

$$P(B|A) = \frac{P(A \text{ and } B)}{P(A)} \quad (4.4b)$$

Independence

$$P(A|B) = P(A) \quad (4.5)$$

General Multiplication Rule

$$P(A \text{ and } B) = P(A|B)P(B) \quad (4.6)$$

Multiplication Rule for Independent Events

$$P(A \text{ and } B) = P(A)P(B) \quad (4.7)$$

Marginal Probability Using the General Multiplication Rule

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \cdots + P(A|B_k)P(B_k)$$

Bayes' Theorem

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \cdots +}$$

Counting Rule 1

$$k^n$$

Counting Rule 2

$$(k_1)(k_2)\ldots(k_n)$$

Counting Rule 3

$$n! = (n)(n-1)\ldots(1)$$

Counting Rule 4: Permutations

$${}_nP_x = \frac{n!}{(n-x)!}$$

Counting Rule 5: Combinations

$${}_nC_x = \frac{n!}{x!(n-x)!}$$

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